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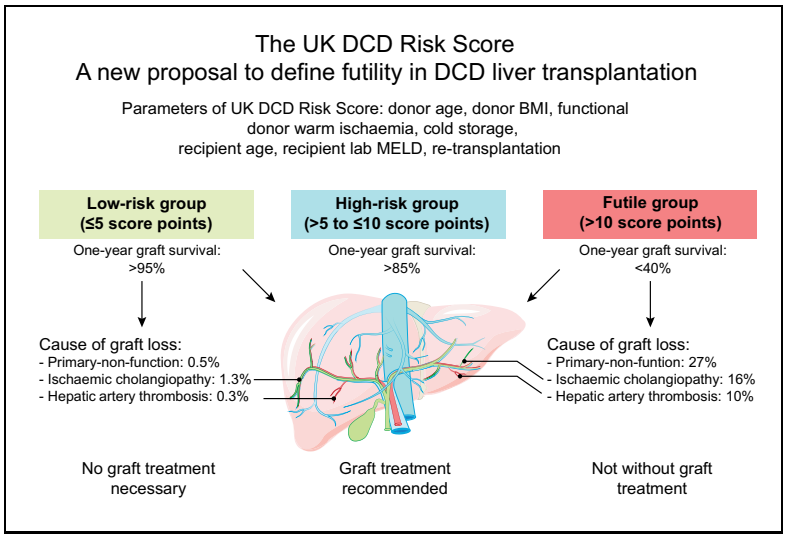
The UK DCD Risk Score: A new proposal to define futility in donation-after-circulatory-death liver transplantation

Graphical abstract



Highlights

- The UK DCD Risk Score best stratifies recipient graft loss after DCD liver transplantation.
- Three risk groups (low risk, high risk and futile) are highly discriminatory for PNF and IC.
- This new model has great potential for improving DCD graft decisions.

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Lay summary

In this study, we provide a new prediction model for graft loss in donation-after-circulatory-death (DCD) liver transplantation. Based on UK national data, the new UK DCD Risk Score involves the following seven clinically relevant risk factors: donor age, donor body mass index, functional donor warm ischaemia, cold storage, recipient age, recipient laboratory model for end-stage liver disease, and retransplantation. Three risk classes were defined: low risk (0–5 points), high risk (6–10 points), and futile (>10 points). This new model stratified best in terms of graft survival compared to other available models. Futile combinations (>10 points) achieved an only very limited 1- and 5-year graft survival of 37% and less than 20%, respectively. In contrast, an excellent graft survival has been shown in low-risk combinations (≤5 points). The new model is easy to calculate at the time of liver acceptance. It may help to decide which risk combination will benefit from additional graft treatment, or which DCD liver should be declined for a certain recipient.

http://dx.doi.org/10.1016/j.jhep.2017.10.034
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The UK DCD Risk Score: A new proposal to define futility in donation-after-circulatory-death liver transplantation

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Background & Aims: Primary non-function and ischaemic cholangiopathy are the most feared complications following donation-after-circulatory-death (DCD) liver transplantation. The aim of this study was to design a new score on risk assessment in liver-transplantation DCD based on donor-and-recipient parameters.

Methods: Using the UK national DCD database, a risk analysis was performed in adult recipients of DCD liver grafts in the UK between 2000 and 2015 (n = 1,153). A new risk score was calculated (UK DCD Risk Score) on the basis of a regression analysis. This is validated using the United Network for Organ Sharing database (n = 1,617) and our own DCD liver-transplant database (n = 315). Finally, the new score was compared with two other available prediction systems: the DCD risk scores from the University of California, Los Angeles and King's College Hospital, London.

Results: The following seven strongest predictors of DCD graft survival were identified: functional donor warm ischaemia, cold ischaemia, recipient model for end-stage liver disease, recipient age, donor age, previous orthotopic liver transplantation, and donor body mass index. A combination of these risk factors (UK DCD risk model) stratified the best recipients in terms of graft survival in the entire UK DCD database, as well as in the United Network for Organ Sharing and in our own DCD population. Importantly, the UK DCD Risk Score significantly predicted graft loss caused by primary non-function or ischaemic cholangiopathy in the futile group (>10 score points). The new prediction model demonstrated a better C statistic of 0.79 compared to the two other available systems (0.71 and 0.64, respectively).

Conclusions: The UK DCD Risk Score is a reliable tool to detect high-risk and futile combinations of donor-and-recipient factors in DCD liver transplantation. It is simple to use and offers a great potential for making better decisions on which DCD graft should be rejected or may benefit from functional assessment and further optimization by machine perfusion.

Lay summary: In this study, we provide a new prediction model for graft loss in donation-after-circulatory-death (DCD) liver transplantation. Based on UK national data, the new UK DCD Risk Score involves the following seven clinically relevant risk factors: donor age, donor body mass index, functional donor warm ischaemia, cold storage, recipient age, recipient laboratory model for end-stage liver disease, and retransplantation. Three risk classes were defined: low risk (0–5 points), high risk (6–10 points), and futile (>10 points). This new model stratified best in terms of graft survival compared to other available models. Futile combinations (>10 points) achieved an only very limited 1- and 5-year graft survival of 37% and less than 20%, respectively. In contrast, an excellent graft survival has been shown in low-risk combinations (≤5 points). The new model is easy to calculate at the time of liver acceptance. It may help to decide which risk combination will benefit from additional graft treatment, or which DCD liver should be declined for a certain recipient.

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Introduction

Driven by organ scarcity and the need to decrease the waiting-list mortality, liver-transplant professionals worldwide have expanded the donor organ pool.[1] In this context, livers from donation-after-circulatory-death (DCD) donors are increasingly used for transplantation, and several countries have implemented DCD programmes.[2–4] However, DCD organs may carry further risk because of the additional donor warm ischaemia. This leads potentially to a higher rate of severe complications, such as primary non-function (PNF) of the graft and intrahepatic ischaemic cholangiopathy (IC).[5] Specific donor-and-recipient risk factors are, therefore, critically evaluated by transplant centres.[6–9] For example, graft cold ischaemia, donor age, and donor body mass index (BMI) or recipient model for end-stage liver disease (MELD) score have all been reported to impact on graft survival.[7] The careful selection of grafts and recipients, therefore, appears decisive for outcome. Data analysis from King's College Hospital (KCH) underlines this fact, showing excellent low rates of IC (2.5%), when the overall risk is low and warm-ischaemia time is kept short with an average of functional donor warm-ischaemia time (fDWIT) of 16 min.[10,11] Such results stand in contrast to other European centres and to previous analyses of US data,[12–14] which report longer donor warm-ischaemia times with higher incidence of IC and graft loss.[15] The combination of too dangerous risk originating from three sources (*e.g.* donor, graft, and recipient) is, however, unclear.

Keywords: Donation after circulatory death; Risk assessment; Graft loss; Morbidity.
Received 23 February 2017; received in revised form 17 October 2017; accepted 25 October 2017; available online 15 November 2017

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The intention of this study was, therefore, to search for an easily applicable score system, based on a few independent donor-and-recipient key factors, with the highest possible accuracy in the prediction of complications and graft loss after DCD liver transplantation.

Patients and methods

Study design and patients

The study was approved by the UK National Health Service institutional review board. The records of all adult (≥18 years) DCD recipients transplanted for chronic liver disease were extracted from a National Health Service Blood and Transplant (NHSBT) analysis and research file from January 2000 through December 2015 (national UK cohort).[16] Two other DCD cohorts were used for validation. First, adult DCD cases were extracted from the United Network for Organ Sharing (UNOS) database from December 1987 through September 2010, after approval by the University of Washington institutional review board (UNOS cohort). Second, we analysed our own institutional DCD database, which included patients transplanted from January 2005 to December 2015 (Birmingham cohort) (Table 1). To reduce the confounding variables, paediatric and partial transplants (split and living-donor liver transplantation), donation after brain death (DBD), and combined liver transplants were excluded. Recipients with lost follow-up were excluded from the analysis. Patients with exception points (*e.g.* hepatocellular cancer within the Milan criteria) were included in the analysis. Importantly, the pretransplant MELD score was calculated without inclusion of those exception points (laboratory MELD). The primary outcome measured for the regression analysis was 1-year graft survival (least missing entries). The final analysis included 1,153 patients. The two validation cohorts, UNOS and Birmingham, consisted of 1,861 and 315 DCD transplant cases, respectively (Table 1).

We defined IC in this analysis on the basis of radiological findings (*e.g.* intrahepatic or hilar biliary strictures and dilatations occurring in the absence of hepatic artery stenosis or hepatic artery thrombosis, portal vein thrombosis, chronic ductopenic rejection, or recurrent primary sclerosing cholangitis). The overall incidence of biliary complications, extrahepatic strictures, IC, and biliary leakages for our Birmingham DCD transplants is described in detail in Table S5.

Statistical analysis

To establish a clinically valid regression model, we chose parameters that are easily available for surgeons at each DCD transplant centre worldwide. Our aim was, therefore, to combine donor, graft, and recipient factors in one practical score system on post-transplant graft survival. All factors should be easily available before transplantation. Accordingly, significant variables with impact on outcome after DCD liver transplantation (with $p < 0.25$) were considered further. These were selected as five metrics and one dichotomous (*e.g.* donor age, fDWIT, cold ischaemia, laboratory MELD score, recipient age, and retransplantation).[17–20] To increase the information on graft quality, donor BMI was also included as a surrogate marker of steatosis and other metabolic disorders.[21] In the UK, fDWIT is defined as time between systolic blood pressure below 50 mmHg and cold aortic organ flush.[22]

For candidate predictors with less than 60% missing values (*e.g.* recipient laboratory MELD score pretransplant [32/1,153; 2.8% missing], fDWIT [464/1,153; 40.2%], and cold ischaemia of the graft [89/1,153; 7.7% missing]), we performed multiple imputation using multivariate normal regression (MVN). We imputed a total of 20 data sets (n = 24,213).[23] The MVN method assumes that the data are missing at random, which appears as a less restrictive assumption than that required by a complete case analysis. MVN has been shown to be valid whether or not all imputed variables follow a normal distribution.[24] In each of the imputed data sets, we used a backward stepwise approach for the multivariable logistic regression with p values ranging between <0.001 and >0.05 as inclusion and exclusion thresholds, respectively. The strongest predictors for a 1-year graft survival after DCD liver transplantation, which were selected in 75–80% of the imputation models, were included in the final multivariable model. Backward elimination is generally preferred as an automated predictor-selection procedure

Table 1. Donor, graft, and recipient characteristics in the three DCD cohorts in the UK, UNOS, and in our centre in Birmingham.

Parameters of DCD liver transplantation	UK n = 1,153	UNOS n = 1,861	Birmingham n = 315
Donor and graft variables:			
Age (years)	49 (35–59)	34 (21–47)	51 (36–62)
Body-Mass-Index (kg/m^2)	25 (23–28)	25.1 (22.0–29.1)	25 (22.7–27.7)
Total warm ischaemia time (min)	27 (22–31)	-	27 (22–32)
Functional warm ischaemia time (fDWIT) (min)	17 (14–20)	14 (9–21)	17 (14–21)
No. of donors with fDWIT >20 min	254 (22%)	364 (19.6)	82 (26)
No. of donors with fDWIT >30 min	40 (3.5%)	124 (6.7)	13 (4.1)
Asystolic warm ischaemia time (min)	13 (11–15)	-	12 (10–14)
Cold ischaemia time (h)	7.1 (6–8.2)	7 (5.3–9)	7 (5.7–8.1)
No. of grafts with CIT >6 h	853 (74%)	1,145 (61.5%)	222 (70.5%)
Recipient variables:			
Age (years)	55 (48–61)	54 (49–60)	58 (51–64)
Body-Mass-Index (kg/m^2)	27 (24–30)	27.4 (23.8–31.2)	27 (24–30)
lab-MELD score at transplantation (points)	15 (11–19)	16.4 (10.8–23.5)	13 (9–17)
UKELD score at transplantation (points)	53 (50–57)	-	53 (49–56)
BAR–Score at transplantation (points)	5 (3–8)	7 (3–10)	5 (3–7)
Follow-up (days)	865 (364–1,704)	600 (160–1,355)	936 (426–1,602)

Data presented as median and IQR for continuous variables or as number and percent for categorical variables. BAR, Balance of Risk Score; BMI, Body-Mass-Index; CIT, cold ischaemia time; DCD, donation after circulatory death; fDWIT: functional donor warm ischaemia time; IQR, interquartile range; MELD, model of end-stage liver disease; UKELD, United Kingdom model of end-stage liver disease; UNOS, United Network for Organ Sharing.

Table 2. Development of the new prediction score based on multivariate regression analysis. The imputed data sets were combined and the point system was developed according to the Framingham risk scheme.[32] For each predictor, the median of all values (Midpoint W_1reference – W_7reference) below and above the threshold has been calculated. The Midpoint for the cohort below each threshold (W_1-7reference) is subtracted from the midpoint of all values above the threshold. The factor β is multiplied with the difference (W_{ij}-$W_i$reference), separately for each factor to develop the score points.[32] BMI, body mass index; CIT, cold ischaemia time; fDWIT: functional donor warm ischaemia time; MELD, model of end-stage liver disease; yr, years; min, minutes; h, hours; β, regression coefficient.

Parameter	Category	Regression coefficient β	*p* value	Reference value Wi (Midpoint)	β x (W_{ij}-$W_i$reference)	Risk score
Donor age	≤60 yr	0.084	0.001	46 (W_1reference)	0	0
	>60 yr			66	1.688	**2**
Donor BMI	≤25 kg/m^2	0.519	0.0001	23 (W_2reference)	0	0
	>25 kg/m^2			28	2.598	**3**
Functional donor warm	≤20 min	0.341	<0.0001	15 (W_3reference)	0	0
ischaemia	>20 to ≤30 min			24	3.069	**3**
	>30 min			32	5.797	**6**
Cold ischaemia time	≤6 h	0.791	0.001	5.5 (W_4reference)	0	0
	>6 h			7.7	1.740	**2**
Recipient age	≤60 yr	0.241	0.0001	52 (W_5reference)	0	0
	>60 yr			64	2.892	**3**
Recipient lab MELD	≤25 points	0.109	0.0001	14 (W_6reference)	0	0
	>25 points			30	1.744	**2**
Retransplantation	No	8.571	<0.0001	0 (W_7reference)	0	0
	Yes			1	8.571	**9**
					Total score points	**0–27**

because it takes correlations among predictors into calculation.[23,25–31] The imputed data sets were combined and the point system was developed to enhance clinical applicability according to the Framingham risk scheme (Table 2).[32] For each predictor, the median of all values below (midpoint W_1 reference–W_7 reference) and above the threshold has been calculated. The midpoint for the cohort below each threshold (W_{1-7} reference) is subtracted from the midpoint of all values above the threshold. The factor β is multiplied with the difference (W_{ij}–$W_{ireference}$), separately for each factor.[32]

Performance and validation of the new prediction model

The performance of the prediction model in the derivation cohort was evaluated by examining measures of discrimination and calibration. Discrimination is the ability of the risk score to differentiate between patients who do and do not experience an event (in our case, the occurrence of graft loss after DCD liver transplantation). This measure was quantified by calculating the area under the receiver-operating-characteristic (ROC) curve (AUC) statistic. Calibration reflects the agreement between predicted probabilities from the model and observed outcomes. We used the Hosmer–Lemeshow test to determine statistically the extent of agreement between the predicted and observed probabilities. We performed an internal validation using a bootstrapping procedure. This approach uses the entire data to develop the prediction model and, in addition, accounts for model overfitting or uncertainty compensating for overoptimism in the final prediction model. The bootstrapping in the current study was performed using 500 bootstrap resamples, each time selecting variables and developing a model within the sample. The new prediction model was compared to available systems from the University of California, Los Angeles (UCLA) (DCD risk score UCLA)[17] and KCH named as DCD risk index (DCD-RI) (Table S4).[33] Furthermore, the model was validated externally in the UNOS database and internally in our own population from Birmingham, where we specifically evaluate the predictive value of our new score, excluding retransplantations. The statistical analysis was performed using SPSS 23.

For further details regarding the materials used, please refer to the CTAT table and Supplementary information.

Results

Are there differences between the three study populations from the UK, UNOS, and Birmingham?

Following eligibility criteria, 1,153 cases were available from the UK database and 1,863 were selected from the UNOS database. Our institutional database included 315 DCD transplants. Median follow-up accumulated to 865 days in the UK, compared to 600 days in the USA and 936 days in our population. No differences were found between the three populations in terms of recipient age (median 55 *vs.* 54 *vs.* 58 years), recipient BMI (27.0 *vs.* 27.4 *vs.* 27.0 kg/m^2), donor BMI (median 25.0 *vs.* 25.1 *vs.* 25 kg/m^2), and cold-ischaemia time (median 7.1 *vs.* 7.0 *vs.* 7.0 h). Major differences (UK data *vs.* US data *vs.* own data) were noted regarding donor age (median 49 *vs.* 34 *vs.* 51 years) and fDWIT (median 17 *vs.* 14 *vs.* 17 min). In the UNOS cohort, centre-dependent variations exist on the time of initiation of donor warm ischaemia. Most US centres, however, define it also as fDWIT, initiated after a specified period of time during sustained hypotension (systolic blood pressure <50 mmHg).[14] The balance-of-risk score at transplantation was slightly higher in UNOS compared to the other cohorts (median 5 *vs.* 7 *vs.* 5 points), corresponding to the slightly higher MELD score in UNOS regions when compared to the European population (median 15 *vs.* 16.4 *vs.* 13 points).[18]

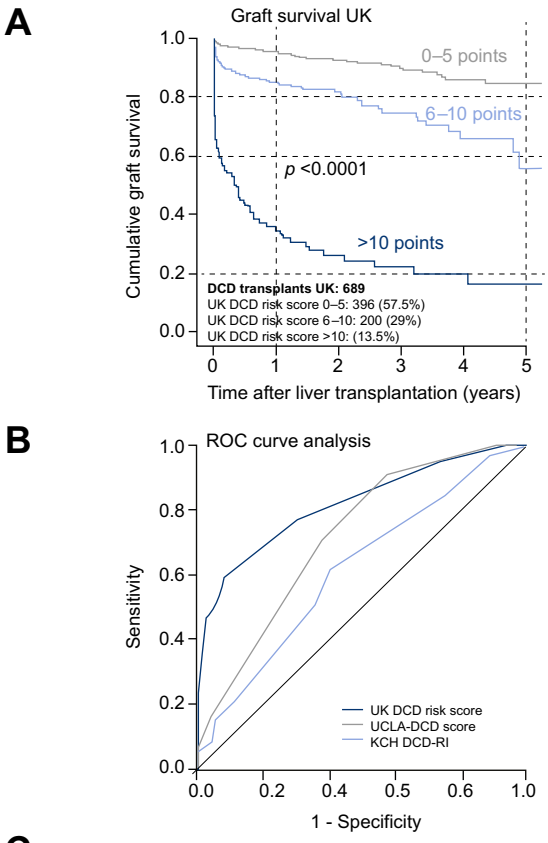
Which are the key prediction factors for graft survival after DCD liver transplantation?

Initial calculations were performed in the large DCD database from the UK and tested, in a next step, in the UNOS and our institutional DCD population. We evaluated 43 candidate predictors. The median number of missing values per candidate was six (interquartile range 2.5–9.5). Five variables were excluded because of more than 60% missing values, which involved aspartate aminotransferase, donor albumin, agonal donor phase, arterial donor pressure, and recipient hospital status at transplant. Among the 38 remaining candidate variables, 28 had complete data and 10 contained <60% missingness and were subjected to the multiple-imputation procedure. The full multivariable prediction model was based on the backward stepwise approach and revealed seven parameters as significant

predictors for graft loss: donor age, BMI, fDWIT, cold-ischaemia time, recipient age, MELD score, and previous liver transplantation (Table 2). The recipient BMI was not identified as an independent predictor. Using the significant regression coefficients, the UK DCD Risk Score was calculated with a range from 0 to 27 points, reflecting an exponential increase in 1-year graft loss. The strongest predictor was retransplantation (0 or 9 points), followed by functional donor warm ischaemia (0, 3, or 6 points), recipient age (0 or 3 points), donor BMI (0 or 3 points), donor age (0 or 2 points), cold-ischaemia time (0 or 2 points), and laboratory MELD (0 or 2 points; Table 2). The internal validation yielded a C statistic of 0.79 (95% CI 0.744–0.839), and the *p* value for the Hosmer–Lemeshow goodness-of-fit test ranged from 0.143 to 0.506 in the 20 imputed data sets, corresponding to a well-calibrated model. The internal validation of the model using the bootstrapping procedure revealed minimal optimism of 0.00005 (95% CI 0.00574–0.064).

How does the newly developed risk score compare to other available prediction systems?

In an attempt to compare the predictability of outcome, C statistics of other scoring systems referring to graft survival were assessed in the UK database. For both available prediction models in the DCD liver transplantation, the UCLA DCD score[17] and the KCH DCD-RI from KCH,[33] the AUCs were 0.71 and 0.64, respectively (Fig. 1). These findings were confirmed by a superior AUC for the new DCD model in our own cohort in Birmingham (AUC = 0.754) compared to the UCLA DCD score (AUC = 0.639) and the KCH DCD-RI (AUC = 0.583) (Fig. S3). Accordingly, with the new UK DCD Risk Score, a more significant decrease in graft survival per increasing score cluster (0–5, 6–10, >10; Fig. S1) has been shown, when compared to both other models. Of note, the balance-of-risk score was also inferior in the prediction of graft survival after DCD liver transplantation in this data set (Fig. S1). In an additional analysis, it has been shown that the cumulative 1-year graft survival per each single score point followed an exponential increase for the new score model above a certain threshold of approximately 10 points (Fig. 2). Both other DCD scores failed to predict cumulative graft loss at each single score point (Fig. 2). Of note, the UK DCD score grouping at this threshold (≤10 *vs.* >10) was better at identifying graft survival than the two other scores, that is, DCD score from UCLA (≤4 *vs.* >4) and DCD-RI from KCH (≤4 *vs.* >4) (Fig. 2). With the newly developed score, very good positive prediction values of 71 and 86.1% in the UK and UNOS, respectively, have been shown. The specificity was 0.95.



C

Scores	Area	SE	Asymptotic sig.	95% CI
UK DCD risk score	0.79	0.024	<0.0001	0.744–0.839
KCH DCD-RI	0.64	0.027	<0.0001	0.562–0.668
UCLA-DCD score	0.71	0.024	<0.0001	0.663–0.757

Fig. 1. Five-year graft survival according to the UK DCD Risk Score and ROC curve analysis (UK cohort). (A) Kaplan–Meier survival plots were developed for graft survival according to the three risk classes of the new score model. Comparisons between groups were made using the log-rank test. Discrimination is the ability of the risk score to differentiate between patients who do and do not experience graft loss. This measure was quantified by calculating the AUC statistic. (B) This is displayed in the ROC curve. (C) The table compares the AUC values of the new model with previously described DCD scores from UCLA and KCH. AUC, area under the receiver-operating-characteristic curve; DCD, donation after circulatory death; DCD-RI, donation-after-circulatory-death risk index; KCH, King's College Hospital; ROC, receiver operating characteristic; SE, standard error; UCLA, University of California, Los Angeles.

Is the new prediction model valid in other DCD cohorts?

The external validation in the UNOS database confirmed the excellent discrimination between the acceptable outcome below the threshold of 10 points and significantly lower graft survival in the futile group with more than 10 UK DCD score points (Figs. 1 and 3). Of note, recipients in the futile group (>10 points) experienced a very high risk for graft loss of more than 60% in 1 year and >80% in 5 years. In addition, candidates in the high-risk group (>5 to 10 score points) lost the graft in 18% and 40% in 1 and 5 years, compared to the lowest-risk group (0–5 points), where 5% and 15% of DCD livers were lost in 1 and 5 years, respectively (Figs. 1 and 3). Importantly, the UK DCD score ranking was also useful in our own cohort, where the strongest predictor, retransplantation, was not available (*p* <0.0001; Fig. 3). Exclusion of cases with a UK DCD Risk Score of more than 10 points from a DCD liver transplant caused by an expected poor prognosis would refer to 13.4% of patients in the UK (93/689) to 14.7% of cases in the USA (237/1,617) and to 11.4% of cases in our population (34/300) (Figs. 1 and 3).

How does the new score system correlate with post-transplant complications and morbidity?

While numerous risk factors have been described previously for DCD liver transplantation, the prediction of IC or also PNF remains unsolved. We were, therefore, particularly interested to identify a combination of key factors with a significant impact on graft loss caused by IC or PNF (Table 3). Donor–recipient constellations with more than 10 score points (futile group) experienced significantly more graft loss caused by PNFs and IC in both large DCD cohorts in the UK and UNOS (*p* <0.0001;

A

Score	Cohort	Score threshold	n	Positive predictive value+	False omission rate*	Delta	C statistic
UK DCD risk score	UK	≤10	596		13.4%	57.5%	0.79
		>10	93	71.0%			
	UNOS	≤10	1,380		22.7%	63.4%	0.72
		>10	237	86.1%			
KC DCD-RI	UK	≤4	501		17.2%	27.1%	0.64
		>4	97	44.3%			
UCLA-DCD score	UK	≤4	572		19.2%	37.9%	0.71
		>4	42	57.1%			

+: Graft loss above the threshold
*: Graft loss below threshold

B Prediction of one year graft loss by different DCD scores

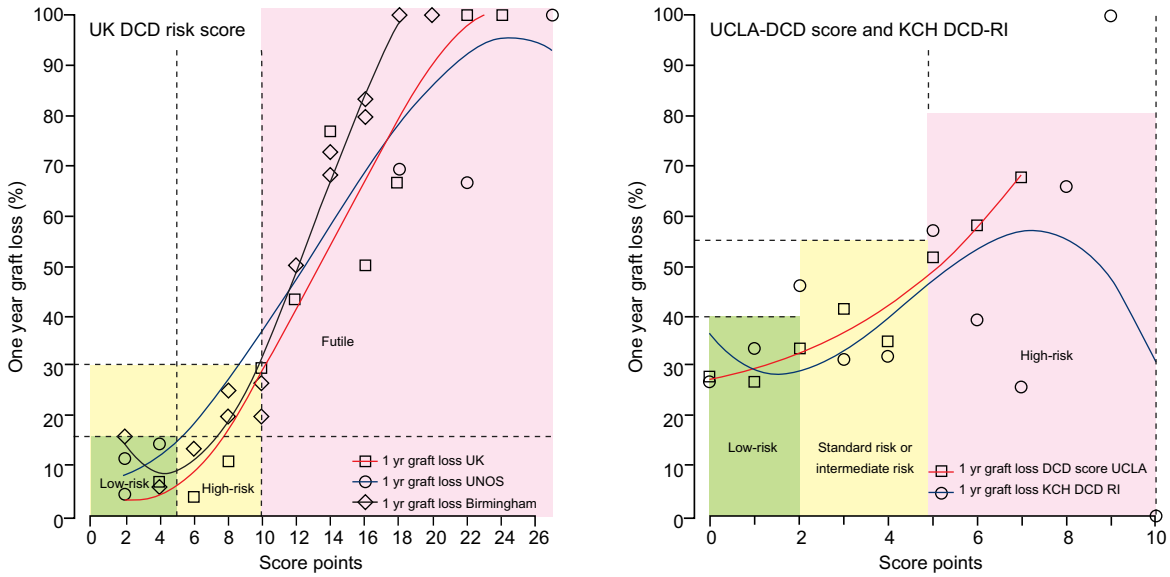


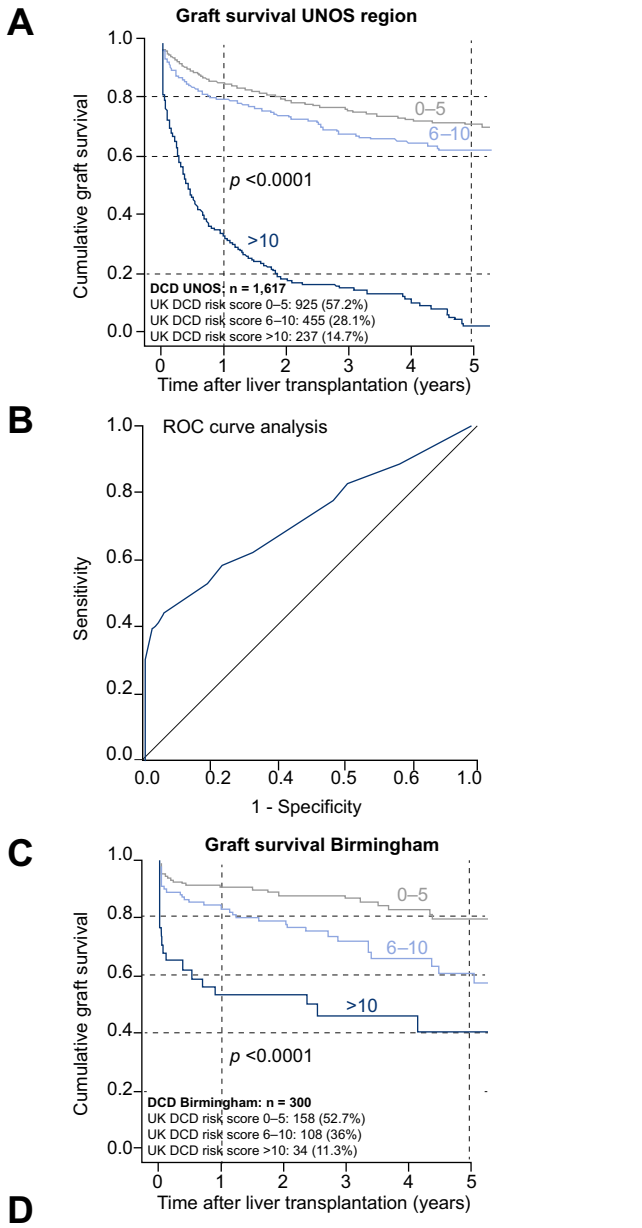
Fig. 2. Prediction of graft loss by the UK DCD Risk Score in all three cohorts, compared to other available DCD risk models. (A) The positive predictive value and false omission rate are shown together with the specificity of three different score models. (B) Survival plots highlight the percentage of grafts, which are lost with increasing score points, for the new prediction model in three different cohorts (UK, UNOS, and Birmingham) compared to two existing models in the UK. DCD, donation after circulatory death; DCD-RI, donation-after-circulatory-death risk index; KCH, King's College Hospital; UCLA, University of California, Los Angeles; UNOS, United Network for Organ Sharing. (This figure appears in colour on the web.)

$p = 0.0001$; Table 4). In addition, recipients in the high-risk and futile groups developed significantly more often vascular complications (*e.g.* hepatic artery thrombosis), leading to graft failure in numerous cases in all cohorts (Table 4). To evaluate liver function and other important post-transplant complications and general morbidity, we analysed our institutional DCD cohort in accordance with the three clusters of the new prediction model. Increasing risk is transmitted by increasing score points resulting in a significantly impaired early liver function (internation normalized ratio, Day 1: 1.5 *vs.* 1.7 *vs.* 2.0, $p = 0.0001$; $p = 0.0005$; Fig. S4) and higher liver-enzyme release (median peak alanine aminotransferase during the first week after transplant (787 *vs.* 1,116 *vs.* 1,906, $p = 0.0004$; $p = 0.0002$). In addition, intensive-care-unit and hospital stay after DCD transplantation increased significantly from low-risk to futile score groups (Fig. S4). The overall post-transplant morbidity was summarised by the comprehensive complication index,[34] where the new DCD prediction score was associated with a significantly increased median comprehensive complication index throughout the three risk groups ($p < 0.0001$; $p = 0.0093$; Fig. S4).

Discussion

Our search for a new score to optimise justice and utility for DCD liver allocation has led to several new findings. First, by combining a few major donor-and-recipient parameters, we developed a simple score system, which best stratifies recipient graft loss after DCD liver transplantation, compared to other recently developed prediction concepts. Second, this score proved to be highly discriminatory for PNF and IC in both the national UK (NHSBT) and USA (UNOS) databases. Third, our calculations represent the largest analysis of DCD liver-transplant recipients. Fourth, the new model correlates well with post-transplant morbidity throughout all three risk clusters.

Risk assessment in liver transplants has been repeatedly addressed by several studies, but is usually limited to the main end points mortality or graft loss. In addition, it appears arbitrary which thresholds and which combinations of parameters in a critically ill patient predict outcome. Currently, the most convincing prediction models for risk analysis in liver transplantation are based on a relatively limited number of key parameters leading to varying score points according to their regression coefficients. The sum of such scores expresses the total risk up to a risk threshold (balance principle).[18,19,35] However, while risk scores have been implemented in DBD liver transplantation, their application in DCD liver transplants is limited, as the DCD population is by far smaller in most countries and parameters depicting higher graft injury are less well defined. Importantly, graft loss caused by PNF and IC is a major and much more relevant problem in DCD liver transplantation in contrast to DBD liver transplantation, and justifies from our point a new effort



D

Scores	Area	SE	Asymptotic sig.	95% CI
UK DCD risk score UNOS region	0.722	0.015	<0.0001	0.693–0.750

Fig. 3. Validation of the UK DCD Risk Score in the UNOS region and Birmingham. (A) Kaplan–Meier survival plots for graft survival were developed to validate the new model in the UNOS DCD cohort. Comparisons between groups were made using the log-rank test. (B) In addition, graft survival in our local DCD liver-transplant cohort in Birmingham is displayed. The AUC statistic, obtained from the UNOS cohort, parallels the values we received from the development cohort in the UK. (C) The table demonstrates a similar AUC in the UNOS cohort, when compared to the UK cohort. AUC, area under the receiver-operating-characteristic curve; DCD, donation after circulatory death; ROC, receiver operating characteristic; UNOS, United Network for Organ Sharing.

to find a formula of their prediction, based on a national analysis.

Minimising the sum of risk is an old principle in surgery, and many transplant professionals automatically aim to allocate DCD livers to low MELD primary liver-transplant candidates.[10,22] This established approach has led to a significantly improved early patient survival after DCD grafting.[12,22] However, the precise impact of different donor/recipient combinations on graft failure is unclear. Two groups have developed models to predict the likelihood of graft survival after DCD liver transplantation.[17,33] In one detailed analysis, Khorsandi *et al.*[33] identified six donor, graft, and recipient factors to predict graft failure after transplantation by the DCD-RI, which combines fDWIT, duration of donor hepatectomy, cold-ischaemia time, recipient MELD, underlying liver disease, and retransplantation. DCD transplant cases are not equally distributed to the three risk clusters of this score from KCH. In contrast to the lowest-risk group, which includes only a very small number of DCD transplant cases (11.5%; 69/598), almost 40% are allocated to the highest-risk group (235/598) and majority of DCD transplantations are classified into the "standard" risk group (50%; 294/598). Based on the number of score points allocated to each case at parameter cut-offs, an inferior prediction of the KCH DCD-RI in terms of graft loss has been shown. For example, one main risk factor in DCD liver transplantation, the fDWIT, contributes to only 1 score point, when ranging above 25 min (53/677 in the UK DCD cohort; 7.8%). According to this score, the majority of DCD livers receive zero points for an already prolonged fDWIT of up to 25 min (624/677; 92.2% of the UK DCD cohort). A similar picture occurs from the parameter, cold-ischaemia time, where the maximum of only 1 parameter point is distributed, when exceeding 10 h. In the UK only, a few DCD grafts experience such long storage times (67/1,153; 5.8%). In contrast, the majority of points (3) are allocated to recipients with HCV infection, alcohol-related liver disease, cryptogenic liver cirrhosis, or Budd–Chiari syndrome as indication for liver transplantation. Such parameters alone may not predict graft loss based on DCD-related complications (PNF and IC), but because of the recurrence of the underlying disease. In addition, 3 score points are allocated to DCD liver recipients with an elevated laboratory MELD of >25 points, which appears only in a limited number of DCD transplantations in the UK (59/691; 8.5%). Thus, the DCD-RI from KCH seems less suitable to stratify post-transplant graft survival when compared to the new UK DCD Risk Score.

Another risk model has been calculated to identify risk combinations, the UCLA DCD score, being a combination of six variables (*e.g.* donor HBV core antibody positivity and functional warm ischaemia [MAP <60 mmHg to flush], cold storage time, recipient MELD, recipient BMI, retransplantation, and underlying liver disease in the recipient).[17] The highest predictive factor with 3 points was HCV positivity combined with hepatocellular carcinoma in this score, which may end up predicting graft loss caused by HCV recurrence rather than PNF or IC. Secondly, the impact of HCV positivity will change in the near future because of the direct-acting antiviral medications.[36] In addition, important predictors of early graft failure (*e.g.* fDWIT) contributed only with 1 score point. An additional score factor, selected by the authors of the UCLA DCD score, published in 2011, was donor HBV core antibody positivity, which is expressed in only very few cases in the UK DCD cohort (14/1,153; 1.2%). Thus, the UCLA DCD score was also inferior in predicting graft survival (Fig. 2). To develop a meaningful prediction model, ideally this should base on clinically relevant score parameters, which are of great importance in all countries with DCD liver transplantation.[35,37,38]

Table 3. Potential combinations of additional key risk factors providing a UK DCD Risk Score ≤10 or >10 points.

	Potential combinations of risk factors	Points
DCD Risk Score ≤10 points	No retransplantation [0] + recipient MELD ≤25 points [0] + recipient age ≤60 yr [0] + cold ischaemia time >6 h [2] + functional donor warm ischaemia >20 to ≤30 min [3] + donor age ≤60 yr [0] + donor BMI ≤25 [0]	5
	Nor [0] + recipient MELD >25 points [2] + recipient age ≤60 yr [0] + cold ischaemia time ≤6 h [0] + functional donor warm ischaemia >30 min [6] + donor age ≤60 yr [0] + donor BMI ≤25 [0]	8
	No retransplantation [0] + recipient MELD >25 points [2] + recipient age >60 yr [3] + cold ischaemia time >6 h [2] + any functional donor warm ischaemia >20 to ≤30 min [3] + donor age ≤60 yr [0] + donor BMI ≤25 [0]	10
	No retransplantation [0]+ any functional donor warm ischaemia up to ≤30 min [0–6] + any donor age [0–2] + any recipient MELD [0–2]	10
	No retransplantation [0]+ functional donor warm ischaemia >20 to ≤30 min [0–3] + any donor age [0–2] + any donor BMI [0–3] + any recipient MELD [0–2]	10
	Retransplantation [9] + recipient MELD ≤25 points [0] + recipient age ≤60 yr [0] + cold ischaemia time ≤6 h [0] + functional donor warm ischaemia ≤20 min [0] + donor age ≤60 yr [0] + donor BMI ≤25 [0]	9
DCD Risk Score >10 points	No retransplantation [0] + recipient MELD >25 points [2] + recipient age >60 yr [3] + cold ischaemia time >6 h [2] + functional donor warm ischaemia >20 to ≤30 min [3] + donor age >60 yr [2] + donor BMI ≤25 [0]	12
	No retransplantation [0] + recipient MELD >25 points [2] + recipient age ≤60 yr [0] + cold ischaemia time >6 h [2] + functional donor warm ischaemia >30 min [6] + donor age >60 yr [2] + donor BMI ≤25 [0]	12
	No retransplantation [0] + recipient MELD >25 points [2] + recipient age >60 yr [3] + cold ischaemia time >6 h [2] + functional donor warm ischaemia >30 min [6] + donor age >60 yr [2] + donor BMI ≤25 [0]	15
	Retransplantation [9] + recipient MELD >25 points [2] + recipient age ≤60 yr [0] + cold ischaemia time ≤6 h [0] + functional donor warm ischaemia ≤20 min [0] + donor age ≤60 yr [0] + donor BMI ≤25 [0]	11
	Retransplantation [9] + recipient MELD >25 points [2] + recipient age ≤60 yr [0] + cold ischaemia time >6 h [2] + functional donor warm ischaemia >20 to ≤30 min [3] + donor age >60 yr [2] + donor BMI ≤25 [0]	19

BMI, body mass index; CIT, cold ischaemia time; DCD, donation after circulatory death; fDWIT, functional donor warm ischaemia time; h, hours; MELD, Model of end stage liver disease; min, minutes; UKELD, United Kingdom model of end-stage liver disease; yr, years.
*Numbers in parenthesis correspond to DCD Risk Score points.

Table 4. Aetiology of graft loss according to the UK DCD Risk Score in the UK, UNOS, and Birmingham DCD cohorts.

Outcome parameter	Overall	Low risk (0–5 points)	High risk (6–10 points)	Futile (>10 points)	*p* value *low risk vs. high risk*	*p* value *high risk vs. futile*
UK	n = 689	n = 396	n = 200	n = 93		
Cause of graft failure						
PNF	29 (4.2%)	2 (0.5%)	2 (1%)	25 (26.9%)	0.6053	*0.0001*
HAT	14 (2%)	1 (0.3%)	4 (2%)	9 (9.7%)	*0.0451*	*0.005*
Biliary (IC)	26 (3.8%)	5 (1.3%)	6 (3%)	15 (16.2%)	0.1942	*0.0001*
One-year graft survival (%)	84.5%	95.5%	85%	37%	*<0.0001*	*<0.0001*
UNOS	n = 1,617	n = 925	n = 455	n = 237		
Cause of graft failure (n/%)						
PNF	87 (5.4%)	21 (2.3%)	12 (2.6%)	54 (22.8%)	0.7090	*0.0001*
HAT	32 (2.0%)	7 (0.7%)	2 (0.4%)	23 (9.7%)	0.7260	*0.0001*
Biliary (IC)	58 (3.6%)	9 (1%)	4 (0.9%)	45 (19%)	1.0	*0.0001*
One-year graft survival (%)	75.3%	85%	80%	33%	*<0.0001*	*<0.0001*
Birmingham	n = 300	n = 158	n = 108	n = 34		
Cause of graft failure (n/%)						
PNF	9 (3%)	2 (1.3%)	2 (1.9%)	5 (14.7%)	1.0	*0.0088*
HAT	17 (5.7%)	7 (4.4%)	6 (5.6%)	4 (11.8%)	0.7746	0.1330
Biliary (IC)	19 (6.3%)	5 (3.2%)	7 (6.5%)	7 (20.6%)	0.2362	*0.0413*
One-year graft survival (%)	84.5%	91%	85%	53%	*<0.0001*	*<0.0001*

DCD, donation after circulatory death; HAT, hepatic artery thrombosis; IC, ischaemic cholangiopathy; PNF, primary non-function; UNOS, United Network for Organ Sharing.

In contrast to the aforementioned calculations, our prediction model discriminated best in terms of graft survival below and above a cut-off of 10 (Table 4; Figs. 1–3). Based on this, donor–recipient combinations with more than 10 points are exposed to a high risk of graft loss caused by clinically relevant and DCD-related complications, such as PNF or IC (Table 4). In addition, the new model significantly correlates with post-transplant morbidity.

Our model has also shortcomings. First, we failed to include graft steatosis in our calculation because of limited donor liver biopsies in the UK. Nevertheless, a correlation of BMI and steatosis has been shown in a recent study,[39] and we have been able to demonstrate donor BMI as an independent predictor.

Second, most DCD recipients in the UK have a low MELD score at the time of transplantation (median laboratory MELD, 15 points). Third, cut-off definitions for higher risk are based on graft survival deterioration with, however, further impact on posttransplant complications. For example, candidates in the high-risk group (6–10 score points) are exposed to significantly higher rate of graft loss caused by IC. This high-risk group would, therefore, benefit from a graft treatment before implantation.[40] In this context, machine-perfusion approaches, such as hypothermic oxygenated perfusion, have been previously shown to reduce reperfusion injury and intrahepatic biliary complications after transplantation of human DCD livers.[41–43] In addition, *ex situ* graft evaluation during normothermic or hypothermic machine liver perfusion might be a useful tool to test liver viability, although better markers of graft function during these approaches are urgently needed.[44–46]

One of the advantages of the UK DCD Risk Score is that it derives from objective factors, readily available at the time of an organ offer, with the exception of warm and cold ischaemia. However, as the liver retrieval starts and the fDWIT becomes available, the score can be recalculated with a reasonably precise prediction of cold ischaemia. At this stage, the score values do suggest a different decision regarding the use of the graft and/or the choice of a different local recipient. In the future, we will calculate the score system at organ offer. When donor/recipient combinations result in more than 10 points, we will either try to find a different recipient to decrease the overall risk or perform graft treatment by machine perfusion before implantation. In addition, future models should also consider the amount of graft steatosis and the impact on the score threshold whether to accept a DCD liver for a certain recipient or not.[47] Although there is an urgent need for more reliable methods of fat quantification, an analysis of graft biopsies provides additional information on the amount of steatosis. It, therefore, helps to decide which DCD liver to accept or how to adapt the preservation technique for a specific risk combination. Because of the fact that important risk parameters (*e.g.* warm-ischaemia time) are frequently missing or incomplete in nationwide databases, further multicentre cohort studies are required to refine existing and new scoring systems. More specifically, an outcome study including DCD transplants from multiple centres in Europe and the USA has been initiated and will enable us to, first, provide another validation of our new model and, second, to demonstrate the effective relation between the number of risk points per DCD case and specific complications in the recipient.

In summary, we believe that the UK DCD Risk Score may be a very practical and new guidance for allocation of a specific organ to a recipient, helping to decide which DCD liver requires processing with a new preservation technology, including machine perfusion. This would potentially help to implement such expensive approaches more specifically.

Conflict of interest

The authors declare that they have nothing to disclose. The data reported here have been supplied partially by the National Health Service Blood and Transplant (NHSBT) and the United Network for Organ Sharing as the contractor for the Organ Procurement and Transplantation Network. The interpretation and reporting of these data are the responsibility of the authors and in no way should be seen as an official policy of or interpretation by the NHSBT or Organ Procurement and Transplantation Network, or the British or US Government.

Please refer to the accompanying ICMJE disclosure forms for further details.

Authors' contributions

AS, PD, PM: study design; AS, MK, IR, RWL: data collection; AS, PD, PM: data analysis; AS, PD, PM: manuscript writing; AS, TP, HM, DFM: structured discussion; and AS, JI, DFM, PD, PM: manuscript revision.

Acknowledgements

The authors are very grateful for the support by Dr. Elisa Allen and Catherine Hopkinson, both working for the Department of Statistics and Clinical Studies, at National Health Service Blood and Transplant, who kindly provided the national donation-after-circulatory-death data, following official approval as stated previously.

Supplementary data

Supplementary data associated with this article can be found, in the online version, at https://doi.org/10.1016/j.jhep.2017.10.034.

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